Decoding Italian Music: **A Text Mining Breakdown via Topic Modeling and Classification**

Mattia Bertolotti (866362)
Gabriele Giorgio (925678)

Text Mining and Search A.Y. 2025/2026

- Dataset creation
- Text Pre-Processing and Representation
- Topic Modelling
- Text Classification
- Conclusion

This project explores the diverse Italian music scene through text analysis. We tackled the complex challenge of lemmatizing Italian lyrics, used topic modeling to extract core themes, and applied text classification to see if an artist's vocabulary actually aligns with their Last.fm genre tags. Finally, we analyzed how these artists approach the historically central theme of love.

A snowball sampling approach for retrieving the artists

Spotify

Last.fm

Genius

Artist lyrics dataset

- artist_id
- artist_name
- genres
- listeners
- playcount
- tracks

Initial corpus exploration and language detection (Lingua)

Text Pre-Processing (spaCy, Stanza)

- Regular Expressions
- Tokenization
- Part-Of-Speech Tagging
- Lemmatization

Text Representation

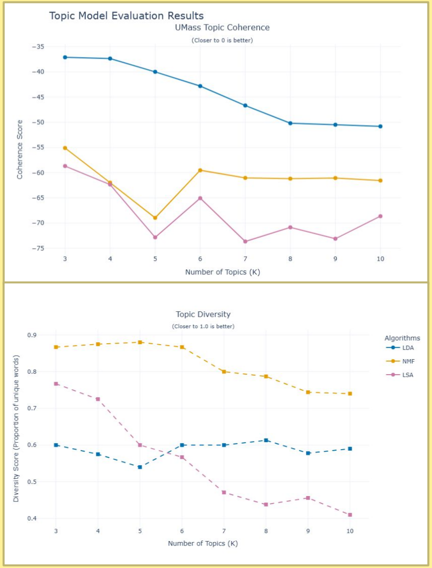
- Bag of World (BoW)
- Term Frequency-Inverse Document Frequency (TF-IDF)

Extract the main topic present in the songs

LDA (Latent Dirichlet Allocation) **BoW**
NMF (Non-Negative Factorization) **TF-IDF**
LSA (Latent Semantic Analysis) **TF-IDF**

Grid Search

- UMass Topic Coherence
- Topic Diversity



Result from NMF (*K* = 6)

Topic	Words and weight per topic									
1	più 2.10	solo 1.03	mai 0.85	ancora 0.81	sempre 0.75	giorno 0.75	notte 0.74	mondo 0.72	qui 0.71	sentire 0.71
2	amore 2.33	amare 0.72	cuore 0.59	bello 0.34	morire 0.29	fiore 0.24	dolce 0.22	donna 0.19	cantare 0.19	grande 0.19
3	avere 2.06	soldo 0.92	stare 0.86	cazzo 0.73	chiamare 0.72	mettere 0.67	prendere 0.61	solo 0.61	amico 0.60	casa 0.58
4	sapere 3.21	dire 1.49	mai 0.85	amare 0.71	stare 0.42	pensare 0.40	solo 0.32	già 0.32	anche 0.30	cosa 0.30
5	volere 2.34	tenere 0.94	nun 0.90	cchiù 0.86	mo 0.73	aggio 0.62	pecché 0.58	ancora 0.46	pure 0.45	vita 0.42
6	andare 3.05	bene 0.93	via 0.73	così 0.53	dove 0.49	mare 0.37	stare 0.35	venire 0.28	lasciare 0.27	dire 0.26

Neural Topic Modeling

VAE with PyTorch

- Reproducibility
- Hyperparameter 6 topics
- Architecture: encoder and decoder
- Loss function
- Early stopping

From song to topic

- Topic word
- Topic confidence
- Useful for text classification
- Misclassified tracks (Tutti, Hubner, ...)

Result from VAE ($K = 6$)

Topic	Words and weight per topic									
1	cchiù 6.873	aggio 6.576	nun 6.290	pecché 6.287	cu 5.535	mo 5.522	quanno 5.328	tenere 5.171	votare 5.015	sordo 4.894
2	gang 5.813	soldo 5.282	troia 4.429	cazzo 4.366	negro 4.193	avere 3.893	bitch 3.783	rapper 3.666	spendere 3.617	chiamare 3.577
3	libero 4.162	terra 3.631	libertà 3.478	sangue 3.451	uomo 3.450	anima 3.417	vivo 3.394	mondo 3.369	fuoco 3.134	nero 3.070
4	ballare 6.105	notte 5.455	luna 5.187	mare 4.559	giù 4.446	estate 4.423	andare 4.027	piovere 4.024	sole 3.861	blu 3.796
5	amore 7.551	amare 6.807	donna 5.431	innamorare 5.290	bello 4.909	volere 4.874	cuore 4.623	dolce 4.563	più 4.518	mai 4.263
6	solo 4.607	mai 4.568	pensare 4.435	volta 4.430	sapere 4.393	perdere 4.359	dire 4.134	stare 4.022	forse 3.859	male 3.795

Comparative Analysis: **NMF vs. Neural VAE**

While NMF mostly clusters generic verbs, the Neural VAE demonstrates vastly superior semantic depth, precisely isolating specific musical subgenres, urban slang, and cultural contexts (like summer hits or rap)

Categorize the songs into predefined classes

Multi-label classification

- Supervised machine learning
- Assigns multiple genre labels to a single textual instance

Genre Pre-Processing

- The heuristic dictionary
- Handling unmapped outliers

Support vector machines via One-vs-Rest

- Restructuring dataset
- Pipeline: *TfidVectorizer* and *OneVsRestClassifier(LinearSVC)*
- Hyperparameter Optimization

	Precision	**Recall**	**F1-Score**	**Support**
Cantautore	0.35	0.75	0.48	36
Classical	0.30	0.90	0.45	30
Electronic	0.36	0.52	0.42	56
Hip-Hop	0.66	0.83	0.74	64
Indie	0.49	0.61	0.54	36
Jazz	0.24	0.67	0.36	30
Pop	0.58	0.67	0.36	30
Rock	0.46	0.42	0.44	73
Macro avg	**0.43**	**0.67**	**0.51**	401

Linear Classifier Chains

- Cascading information
- Pipeline: *TfidVectorizer* and *ClassifierChain(LinearSVC)*
- Hyperparameter Optimization

	Precision	Recall	F1-Score	Support
Cantautore	0.39	0.42	0.41	36
Classical	0.29	0.77	0.42	30
Electronic	0.31	0.07	0.12	56
Hip-Hop	0.65	0.81	0.72	64
Indie	0.73	0.53	0.761	36
Jazz	0.40	0.13	0.20	30
Pop	0.61	0.63	0.62	76
Rock	0.44	0.32	0.37	73
Macro avg	**0.48**	**0.46**	**0.43**	401

Tree-Based Classifier Chains

- Gradient boosting
- Pipeline: TfidVectorizer and ClassifierChain(XGBClassifier)
- Hyperparameter Optimization

	Precision	Recall	F1-Score	Support
Cantautore	0.46	0.17	0.24	36
Classical	0.64	0.23	0.34	30
Electronic	1.00	0.04	0.07	56
Hip-Hop	0.78	0.72	0.75	64
Indie	1.00	0.11	0.20	36
Jazz	0.50	0.03	0.06	30
Pop	0.61	0.58	0.59	76
Rock	0.31	0.42	0.36	73
Macro avg	**0.66**	**0.29**	**0.33**	401

Transformer BERT

- Dataset flattening
- Tokenization
- Anti-Overfitting defenses
- Algorithmic compensation
- Post-Training calibration

	Precision	Recall	F1-Score	Support
Cantautore	0.38	0.71	0.50	625
Classical	0.46	0.50	0.48	141
Electronic	0.22	0.43	0.29	343
Hip-Hop	0.74	0.79	0.77	1058
Indie	0.35	0.64	0.45	513
Jazz	0.16	0.38	0.23	235
Pop	0.61	0.74	0.67	1165
Rock	0.33	0.65	0.44	768
Macro avg	**0.41**	**0.61**	**0.48**	4848

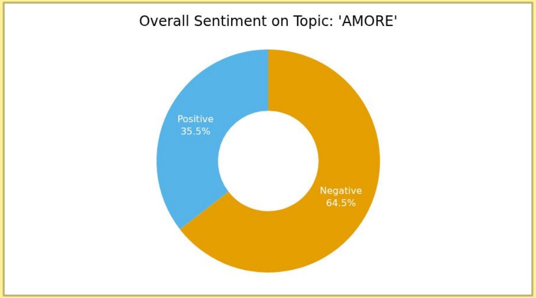
Comparative Analysis: **ML vs. BERT**

- **Precision**: Accuracy of predicted positive cases.
- **Recall**: Finding all the actual positives.
- **F1-Score**: Balance between precision and recall.

Based on neural network topic modeling, this section isolates tracks categorized under 'Love' (Amore) to analyze their sentiment and lyrical themes (FEEL-IT)

Theme of Love

- **A Melancholic Majority**: 64.5% as negative sentiment
- **The Joyful Minority**: 35.5% as positive sentiment



This research tried to decoded the linguistic and thematic landscape of over 20,000 Italian song lyrics

Pre-Processing

- Traditional tools are insufficient
- Stanza's neural pipeline achieve the morphological precision

Topic Modeling

- NMF: Broad dominant themes
- VAE: Superior semantic depth

Text Classification

- UmBERTo: Mainstream genres
- SVM: Minority genres

Thank you for your time

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